Customer Churn Risk Profiler for the Telecom Industry

Stage 1 Project Proposal

Project Title	Customer Churn Risk Profiler for the Telecom Industry
Domain	Marketing & Brand Management (Customer Retention Analytics)
Working Group	G2
Group Number	14
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1. Business Problem Statement & Objectives

Business context. Telecom is a subscription business, so losing a customer means losing future recurring revenue. Retention teams have limited budgets and need to know which customers are more likely to leave and which characteristics are associated with that risk. This project simulates a telecom operator and builds a Python-based churn risk profiler on a **synthetically generated** customer dataset.

Target audience. A telecom retention or marketing manager who must prioritize customers for retention activity and justify where the budget goes.

Core question. Which customers appear to be at higher risk of churn, which factors are associated with that risk, and how can the business respond?

Analytical goals.

1. **Measure** overall churn and compare churn across customer segments (contract, tenure, charges, satisfaction, support tickets, autopay).
2. **Estimate** a churn probability for every customer with an explainable model and assign each customer to a Low, Medium or High risk band.
3. **Identify** the top three churn-associated drivers from the data and model evidence.
4. **Convert** the findings into specific, segment-based retention recommendations.

All findings will be described as statistical associations within the simulated data, not as proven causes of churn.

2. Synthetic Dataset Design & Schema

Data generation strategy. We will generate approximately **5,000 customer records** in Python using random and numpy, with a **fixed random seed** for reproducibility. The data is independently generated. The IBM Telco Customer Churn sample will be used only as an external reference to sanity-check variable selection and terminology. **No IBM** rows, churn rates or results will be copied or presented as ours. The dataset will be saved as a **CSV file**.

Data dictionary

Column	Type	Generation rule
customer_id	String	CUST00001 to CUST05000
age	Integer	Normal (mean ≈ 38, SD ≈ 12), clipped to 18–75
city_tier	Categorical	Metro 35%, Tier-2 40%, Tier-3 25%
tenure_months	Integer	1–72 months

contract_type	Categorical	Month-to-month / One year / Two year; longer-tenure customers have a higher share of one- and two-year contracts
internet_service	Categorical	Mobile only 30%, Broadband 30%, Fiber 40%
monthly_charges	Float (₹)	Service-dependent base plus normal noise, clipped to ₹200–₹1,500
support_tickets	Integer	Poisson (mean ≈ 1.5), capped at 8
satisfaction_score	Integer	1–5, mildly related to support tickets (more tickets tend to lower scores)
autopay	Categorical	Yes ≈ 55% / No ≈ 45%
payment_delay_days	Integer	Mostly 0 for autopay customers; more frequent delays for non-autopay customers; maximum 30
usage_hours	Float	Normal (mean ≈ 120, SD ≈ 40), clipped to 10–300
churn	Integer	0 = retained, 1 = churned (generated as described below)

Business logic built into the data.

Churn will be generated in three steps. A weighted **risk score** is calculated from customer attributes. The score is converted to a probability with the **sigmoid function**. A random 0/1 outcome is then drawn from that probability, so the data keeps realistic noise and is not deterministic. The intended relationships are:

- Shorter tenure → higher churn risk.
- Month-to-month contracts → higher risk than one- or two-year contracts.
- Higher monthly charges → moderately higher risk.
- More support tickets → higher risk.
- Lower satisfaction → higher risk.
- More payment-delay days, and no autopay (slightly) → higher risk.
- Usage has only a small effect.
- Age and city tier are **not** major components of the formula by design.

The intercept will be calibrated toward an overall churn rate of about 20%. This is a DESIGN TARGET only. It is not an expected empirical result or a confirmed churn rate. The actual rate will be measured after generation. The churn column and the hidden risk formula will never be used as model inputs, to avoid data leakage.

3. Proposed Technical Methodology

Tools. Python in a notebook (Jupyter/Google Colab), using pandas, numpy, random, matplotlib, seaborn and scikit-learn. The code will be organized into simple, clearly named functions for generation, validation, analysis, visualization and scoring. The project will not use object-oriented programming, SQL, deep learning or web frameworks.

Planned workflow

1. **Data generation and file handling.** Generate the records, build a pandas DataFrame and save it as a CSV file. The notebook then reloads the CSV for all analysis.
2. **Validation and error handling.** Check row count, unique customer IDs, missing values, data types, valid ranges, allowed categories and the churn proportion. Basic exception handling will cover a missing file, an empty dataset, invalid categories, division by zero in rate calculations, and out-of-range manual inputs.
3. **pandas analysis.** Inspect the data with head(), info(), describe() and value_counts(). Create derived analysis columns such as tenure, charge, satisfaction and support-level groups.
4. **KPIs and segment analysis.** Calculate total customers, churned customers and overall churn rate. Compare average tenure, charges, satisfaction and tickets by churn status. Compare churn rates by contract type, tenure band (0–12, 13–24, 25–48 and 49–72 months), charge band, satisfaction level, support-ticket level and autopay status.
5. **Visualizations.** Eight charts are planned. Each will have a title, labelled axes and a short business interpretation.
   - *Six analytical charts:* churn by contract type, churn by tenure group, monthly charges versus churn, satisfaction versus churn, support tickets versus churn, and a numeric correlation heatmap.

- *Churn-risk distribution chart:* shows how customers are spread across the Low, Medium and High risk bands and how the predicted probabilities are distributed.
- *Top-driver visualization:* compares the strength of the main factors associated with churn, based on model evidence and segment analysis.

6. **Logistic Regression.** Churn is a binary outcome, and Logistic Regression gives an explainable probability, so we will use it. Categorical variables will be encoded with a simple method, and the data will be split into training and test sets. We will evaluate with accuracy, precision, recall and a confusion matrix.
7. **Customer-level churn probability and risk classification.** Each customer will receive a predicted churn probability. Customers will be classified into Low, Medium and High risk bands using project-defined probability thresholds. The final cut-offs will be reviewed after examining the model's probability distribution and documented in the final analysis.
8. **Top three churn drivers.** We will compare segment churn rates with model coefficients and select three factors that are both analytically meaningful and actionable. **The three drivers will be selected only after the actual analysis is run.** They will not be decided in advance. Because contract type is tied to tenure and autopay to payment delay, coefficients will be interpreted with caution.
9. **Managerial recommendations.** Each recommendation will state the observed pattern, the affected segment and the proposed business action, based only on the actual results.

Expected output. A notebook that generates and validates the dataset, produces KPIs and charts, and scores customers. It will also let a user enter a hypothetical customer to see an estimated churn probability and risk band.

4. Timeline & Responsibilities

Phase	Milestone	Definition of done
1	Stage 1 proposal	Proposal submitted
2	Dataset generator	Generator runs; CSV created
3	Validation	All validation checks pass
4	Analytics	KPIs and segment analysis complete
5	Visualization	Planned charts completed with interpretations
6	Model	Logistic Regression trained; actual metrics recorded
7	Risk bands, drivers and recommendations	Cut-offs reviewed; top three drivers and recommendations finalized from actual results
8	Final report	Report completed using only actual results
9	Presentation and viva preparation	Slides done; all members rehearsed

Proposed responsibilities. Roles have not yet been formally assigned, so the split below is a proposal. It is primarily based on each members strengths and roughly based on the group's comfort with Python as of now.

Member	Proposed responsibilities
R. Ruchita Gowri	Data generation, dataset validation, core Logistic Regression model, and menu-driven interface
Krrish Bahuguna	Visualizations, chart interpretation, and managerial recommendations
Rishit Hotchandani	pandas analysis, KPIs, segmentation, and model evaluation
Ashi Beniwal	Detailed Data checks, documentation, and report and presentation slide support

Shared understanding. Responsibilities are divided for efficiency, not to limit anyone's knowledge. All four members will walk through the full pipeline together, from data generation to recommendations, and will be prepared to explain every major function, chart and decision in the viva. Members will review each other's sections, and more experienced members will help explain their parts to the others. This is so that no member depends on one teammate to defend the project.